Finishing the Network Layer & Transitioning to the Link Layer

Data vs. Control Plane & Introduction to Link Layer Services

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Recap: Transitioning Down the Stack

- In the previous two chapters, we learned that the network layer provides a communication service between any two network hosts.
- Between the two hosts, datagrams travel over a series of communication links, some wired and some wireless, starting at the source host, passing through a series of packet switches (switches and routers) and ending at the destination host.
- While the Network Layer computes the end-to-end path (the Control Plane) and forwards packets locally (the Data Plane), it must rely on the protocol below it to actually cross the physical wires.

The Travel Agent Analogy

Consider a travel agent who is planning a trip for a tourist traveling from Princeton, New Jersey, to Lausanne, Switzerland.

- The travel agent decides that it is most convenient for the tourist to take a limousine from Princeton to JFK airport, then a plane from JFK airport to Geneva's airport, and finally a train from Geneva's airport to Lausanne's train station.
- **The tourist** is a datagram.
- **Each transportation segment** is a link.
- **The transportation mode** (limousine, plane, train) is a link-layer protocol.
- **The travel agent** is a routing protocol.

Link Layer Terminology

As we transition to Layer 2, we must use precise terminology:

- **Nodes:** We refer to any device that runs a link-layer (i.e., layer 2) protocol as a node. Nodes include hosts, routers, switches, and WiFi access points.
- **Links:** We will also refer to the communication channels that connect adjacent nodes along the communication path as links.
- **Frames:** Over a given link, a transmitting node encapsulates the datagram in a link-layer frame and transmits the frame into the link.

Link Layer Services (Part 1)

Although the basic service of any link layer is to move a datagram from one node to an adjacent node over a single communication link, possible services offered include:

- **Framing:** Almost all link-layer protocols encapsulate each network-layer datagram within a link-layer frame before transmission over the link. A frame consists of a data field, in which the network-layer datagram is inserted, and a number of header fields.
- **Link Access:** A medium access control (MAC) protocol specifies the rules by which a frame is transmitted onto the link. This is critical when multiple nodes share a single broadcast link.

Link Layer Services (Part 2)

- **Reliable Delivery:** When a link-layer protocol provides reliable delivery service, it guarantees to move each network-layer datagram across the link without error. This is often used for links that are prone to high error rates, such as a wireless link. It can be considered an unnecessary overhead for low bit-error links, including fiber and coax.
- **Error Detection and Correction:** Bit errors are introduced by signal attenuation and electromagnetic noise. Error detection in the link layer is usually more sophisticated than the Internet checksum and is implemented in hardware. Error correction allows a receiver to not only detect bit errors but determine exactly where they occurred and correct them.

Where is the Link Layer Implemented?

The link layer is the place in the protocol stack where software meets hardware.

- **Hardware (Network Adapter):** For the most part, the link layer is implemented on a chip called the network adapter, also sometimes known as a network interface controller (NIC). Much of a link-layer controller's functionality (framing, link access, error detection) is implemented in hardware.
- **Software (CPU):** Part of the link layer is implemented in software that runs on the host's CPU. This software responds to controller interrupts, handles error conditions, and passes datagrams up to the network layer.

Multiple Access Links

There are two types of link-layer channels: point-to-point links and broadcast links.

- **The Cocktail Party Problem:** A human analogy for a broadcast channel is a cocktail party, where many people gather in a large room to talk and listen.
- **Collisions:** Because all nodes are capable of transmitting frames, more than two nodes can transmit frames at the same time. When this happens, all of the nodes receive multiple frames at the same time; that is, the transmitted frames collide at all of the receivers.
- **The Solution:** We use multiple access protocols to coordinate the transmissions of the active nodes.

Classes of Multiple Access Protocols

We can classify just about any multiple access protocol as belonging to one of three categories:

1. **Channel Partitioning Protocols:** Techniques like time-division multiplexing (TDM) and frequency-division multiplexing (FDM) that partition a broadcast channel's bandwidth among all nodes.
2. **Random Access Protocols:** A transmitting node always transmits at the full rate of the channel. When there is a collision, each node repeatedly retransmits its frame until it gets through without a collision, waiting a random delay before retransmitting.
3. **Taking-Turns Protocols:** Protocols like polling (using a master node) or token-passing (exchanging a special-purpose frame) to ensure nodes take turns.

Random Access: CSMA/CD

Carrier sense multiple access with collision detection (CSMA/CD) mimics polite human conversation:

- **Carrier Sensing:** Listen before speaking. A node listens to the channel before transmitting.
- **Collision Detection:** If someone else begins talking at the same time, stop talking. A transmitting node listens to the channel while it is transmitting. If it detects an interfering frame, it stops transmitting.
- **Binary Exponential Backoff:** When transmitting a frame that has experienced collisions, a node chooses a random wait time. The size of the sets from which the wait time is chosen grows exponentially with the number of collisions.

MAC Addressing

- It is not hosts and routers that have link-layer addresses but rather their adapters (that is, network interfaces) that have link-layer addresses.
- A link-layer address is variously called a LAN address, a physical address, or a MAC address.
- For most LANs (including Ethernet and 802.11 wireless LANs), the MAC address is 6 bytes long, giving 2^{48} possible MAC addresses.
- **Structure:** An adapter's MAC address has a flat structure and doesn't change no matter where the adapter goes. This is analogous to a social security number, whereas an IP address is hierarchical, analogous to a postal address.

Address Resolution Protocol (ARP)

Because there are both network-layer addresses (IP addresses) and link-layer addresses (MAC addresses), there is a need to translate between them.

- For the Internet, this is the job of the Address Resolution Protocol (ARP).
- An ARP module in the sending host takes any IP address on the same LAN as input, and returns the corresponding MAC address.
- Unlike DNS which resolves host names anywhere in the Internet, ARP resolves IP addresses only for hosts and router interfaces on the same subnet.
- Each host and router has an ARP table in its memory, which contains mappings of IP addresses to MAC addresses.

Ethernet

Today, Ethernet is by far the most prevalent wired LAN technology.

- **Frame Structure:** Consists of a Preamble, Destination MAC, Source MAC, Type, Data (payload), and CRC.
- **Service Model:** All Ethernet technologies provide connectionless service to the network layer (no handshaking).
- **Reliability:** Ethernet technologies provide an unreliable service to the network layer. When a frame fails the CRC check, the adapter simply discards the frame without sending a NAK.

Link-Layer Switches

The role of the switch is to receive incoming link-layer frames and forward them onto outgoing links.

- **Transparent:** A switch does this job transparently, without the host or router having to explicitly address the frame to the intervening switch.
- **Forwarding based on MAC:** A switch is a store-and-forward packet switch, but it is fundamentally different from a router in that it forwards packets using MAC addresses.
- **Self-Learning:** Switches are plug-and-play devices. The switch table is built automatically, dynamically, and autonomously by observing the source MAC addresses of incoming frames.